Input, Process, Output

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Helpful Classroom Sorter

TEACHER GUIDE · 1 OF 9

Input, Process, Output

> Prepare Input, Process, Output: check the target, prerequisites, resources and adaptable teaching schedule.

Teach the learning cycle

Primary target: **Trace** information through a system without confusing information with a physical action

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Compare the cases, Words that unlock the idea and Follow a worked example; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.

> **Resources:** Use Find and repair a mistake, Practise together, Run the investigation and Build your understanding; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project, Show what you know, Study a complete case and Try a new case independently. Allow extra time for the full task set. Use a coding lab only when this week actually provides one; Your lesson route is the planning section.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Retrieval answer guidance

AI or Not?

Without opening the old lesson, explain one key idea from AI or Not?. Give an example and a mistake that the idea helps you avoid.

Look for: Distinguish an observed result from an explanation of how it was produced

Reference example, not the only acceptable wording: **Observation**s include that the door opened when the visitor approached and that its label says intelligent entrance.

A useful question asks whether a written sensor threshold, a model fitted on examples or a person controls opening. The label is an observed marketing claim, not **mechanism evidence**.

Use the student module and workbook section with this exact week title. The worked case, independent question and answers below form one aligned lesson sequence.

What the learner must demonstrate

> A library kiosk receives 'mystery', filters a catalogue by genre and displays three titles. No speaker is attached. Draw its information path. A visitor says it read three books aloud. Which stage contradicts that claim?

Concept to explain

A system transforms information; it does not simply 'know.'

Term	Meaning
input	information entering a system
process	the steps that transform information
output	the result produced by the system
trace	follow information through each stage

Prepare the class

Use the supplied scenario, student illustrations and blank paper. For numeric work, offer counters or a calculator after learners show the setup. Fictional records are supplied; personal data and paid accounts are unnecessary.

Alternative compact allocation

> This compact option is not the full-week schedule. Use the main teaching cycle and allow extra sessions for the investigation, independent work and project.

Allow 80 minutes: recall 8; explanation 12; worked demonstration 15; guided practice 20; independent assessment 15; project and reflection 10. Extend the lab into a second session when needed.

Preparation for the connected explanation

Teach **input** and **output** as relative to a stage. Avoid claiming that **representation**s are inherently truthful or complete.

This lesson introduces **information loss** through counterexamples, without requiring formal information theory.

The two-originals method is the important reasoning move: if two source records map to one stored result but imply different answers, the stored result cannot decide that question alone.

Readiness answer

The fitted model produced the delay estimate; the written greater-than-10 rule selected the colour. A caretaker or driver decision is a separate action.

Catalogue records A science 80, B science 150 and C mystery 90 written on cards

The information-path diagram and blank arrow diagrams

Paper for two four-number lists; no devices or accounts are needed

End goal for the pictorial case

By the end, I can **trace** an information path and name what a summary loses.

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

Explain the mechanism

> Model Input, Process, Output using the supplied case, labelled mechanism and reasoning. Make the scope of each conclusion explicit.

Follow the information instead of imagining magic inside the machine.

1. Input — information entering

INPUT songs played, liked, or skipped

2. Process — the deciding step

PROCESS compare clues and rank possible songs

3. Output — the result

OUTPUT a suggested song or a not-sure result

Key idea

CORE IDEA A system transforms information; it does not simply 'know.'

How the mechanism works

UNDER THE HOOD **Input**, **process**, and **output** are functional roles. The **process** may contain human choices, fixed rules, calculations, retrieval, or a **learned model**.

Naming the three stages prevents magical explanations of computer behavior.

Draw a fresh example with all three stages:

Every step connects a claim to observable **evidence**.

The case to investigate

CASE A music helper receives listening choices, compares them with a ranking **process**, and suggests a song. The suggestion appears only after information has moved through the system.

Step 1

STEP 1 — OBSERVE songs played, liked, or skipped

Step 2

STEP 2 — **TRACE** compare clues and rank possible songs

Step 3

STEP 3 — CONCLUDE a suggested song or a not-sure result

Model answer

MODEL REASONING Start with what enters, name the actual transformation rather than saying the system thinks, and identify the result. If the **process** is unknown, state what **evidence** is missing instead of inventing a **mechanism**.

A claim to avoid

DO NOT OVERCLAIM The **output** is not proof that the system understood the **input** as a person would.

What new **evidence** could change this answer?

Concrete teaching case

The teaching case

A playlist helper receives a mood, matches song tags and displays three titles.

Name **input**, **process** and **output**. Is displaying titles the same as playing audio?

Exact explanation to model

Why the result follows

Input is mood; **process** is tag matching; **output** is titles. Audio playback is another step. Matching can use fixed rules; the description alone does not establish machine learning.

Explain the added worked example

Explain the added worked example · 1

A library kiosk asks Sam for a genre. Sam selects mystery. The kiosk compares that preference with catalogue tags and displays three book titles. Sam says the kiosk has read three books aloud.

That conclusion adds an action that the description never supplied: displaying a title is different from producing spoken audio.

Explain the added worked example · 2

An information path names what enters each stage, what happens there and what leaves it. Treat **input** and **output** as roles relative to a stage.

A list of titles is the **output** of the search, but it could become the **input** to a separate text-to-speech stage. A component can therefore receive the **output** of another component without doing the same job.

The kiosk's genre-matching operation can be implemented with a **written rule**. Nothing in this description establishes that a model was trained.

This connects to Week 1: tracing information tells us what a **process** needs to do, while **mechanism evidence** tells us how it does it.

Explain the added worked example · 3

Notice the difference between a title, a description and an entire book. All three can be text, but they represent different amounts of information.

Even a speech stage receiving three titles can speak only those supplied titles; it has not thereby read the full books. Always identify the content of an **input**, not just its data type.

A recommendation is also different from a loan. Selecting a title does not reserve a copy unless an authorised reservation step exists.

Keeping these distinctions clear prevents us from crediting a system with actions it did not take.

Key terms

Term	Meaning
Representation	A way of recording information so that it can be stored, communicated or processed
Input	Information supplied to a particular process
Process	An operation that changes, selects or interprets information
Output	The result produced by a process
Information loss	Details that cannot be recovered from the retained representation

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Model the reasoning aloud · 4

Consider two catalogue records. Book A has 90 pages and is about a child solving a seaside mystery. Book B has 90 pages and is about a lost spaceship. If we keep only page count, both become the number 90.

The number is useful for a length limit, but it cannot tell us which story is a mystery. Information that was never retained cannot be recovered just by calculating harder.

Representation kept	Question it can help answer	Question it cannot settle alone
Page count 90	Is the book under 100 pages?	Is its story a mystery?
Genre tag mystery	Does it match a mystery request?	Will this particular reader enjoy it?
Short summary	What does the description say happens?	What exact sentence occurs on page 42?

Model the reasoning aloud · 5

The phrase cannot settle alone matters. A genre tag might be inaccurate, and a reader might enjoy an unfamiliar genre. A **representation** is useful only in relation to a question and its limits.

More detail is not automatically better: unnecessary personal information can add risk without helping the task.

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Input, Process, Output

A fictional classroom cupboard records coloured pencils, then reports just a total.

The facts we will use

> **Supplied facts**

Box A has 2 red and 4 blue pencils. Box B has 5 red and 1 blue.

Supplied facts

The **process** adds colour counts and stores only the total, not the two counts.

AI ACADEMY / AI-1/2

Different colour counts, same total

	Red	Blue	Total
Box A	2	4	6
Box B	5	1	6

Read a cell using both its row and column labels.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-1/2

Input, Process, Output

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Read colours

A: red 2, blue 4. B: red 5, blue 1.



2. Add counts

A: 2+4=6. B: 5+1=6.



3. Store total

Both records become total 6.



4. Ask a new question

The saved totals cannot tell which box has more red pencils.

CHANGE ONE THING

Store red count alongside total.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Read colours	A: red 2, blue 4. B: red 5, blue 1.
2. Add counts	A: 2+4=6. B: 5+1=6.
3. Store total	Both records become total 6.
4. Ask a new question	The saved totals cannot tell which box has more red pencils.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

Adding preserves the number of pencils but discards how that number was divided between colours. Both distinct **input**s lead to the same stored **output**.

Working backwards from 6 therefore has several possible answers; it is not an instruction to divide equally.

What this example does not establish

Keep the stated limit

The two-colour assumption matters. Extra colours would prevent inferring blue from total and red alone.

Change one thing in the pictured case

Change one thing

Change: Store red count alongside total.

Predicted result

Prediction: The records become (6,2) and (6,5).

Why the result follows

Reason: Blue count can now be recovered by total minus red, assuming these are the only colours.

Ask while showing the picture

Ask while showing the picture · 6

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a **representation**; it does not document execution of a real AI model.

Demonstrate and guide practice

> Demonstrate Input, Process, Output, reduce support deliberately and check individual reasoning before discussing answers.

Board sequence

1. Put the facts on the board

Write the given facts: A playlist helper receives a mood, matches song tags and displays three titles.

2. State the decision to explain

Write the question to explain: Name **input**, **process** and **output**. Is displaying titles the same as playing audio?

3. Reveal the worked reasoning

Reveal after **prediction**s: **Input** is mood; **process** is tag matching; **output** is titles. Audio playback is another step. Matching can use fixed rules; the description alone does not establish machine learning.

Hold back the independent task

Keep for an independent attempt: A classroom fan receives a temperature reading, compares it with 25 degrees C, and turns on. Name the **input**, **process** and **output**. What should happen if the reading is missing?

Classroom dialogue

Ask the class

Ask: Which facts are given, and which would we have to assume? Learners mark relevant details before discussing answers.

Language to model

Say: Predict first. Show the rule, information path or calculation that would produce the result. A correct guess is not yet an explanation.

Reveal reasoning one step at a time. Explain why a tempting alternative fails. Write units and denominators for arithmetic, **trace** conditions for decisions, and identify supporting sources for claims.

In pairs, one learner solves while the other checks **evidence**; exchange roles. Remove the worked answer before independent assessment.

Model these guided questions

Visitor schedules can be three at 09:00 or one each at 09:00, 10:00 and 11:00. The total does not reveal timing or busiest hour.

Displaying a book title produces text on a screen; speaking needs audio generation, reserving needs an authorised availability update and reading the whole book needs its full content.

Case-specific teaching sequence

Stage	Teacher action
Session 1 start 0 to 7 minutes	Retrieve the bus stages from Week 1. Ask which result becomes a later input.
7 to 20 minutes	Read the kiosk case and trace the first diagram row. Cover later rows, then reveal them as additional operations rather than assumed capabilities.

Stage	Teacher action
20 to 33 minutes	Use the two 90-page book descriptions. Ask which question survives when only 90 is kept. Model one counterexample.
33 to 45 minutes	Pairs complete W1 and the visitor example. Require exact input fields and outputs.
Session 2 start 0 to 8 minutes	Ask whether a largest temperature proves a rising sequence. Have two students construct alternative sequences.
8 to 23 minutes	Complete W2 and W3. Separate current records, exact source text and summary tags.
23 to 38 minutes	Collect W4 independently. Give no example list until the attempt is saved.
38 to 45 minutes	Use W5 to connect information loss with stage diagnosis. Review the exit explanation and preview measurement.

Reduce help deliberately

Completion practice

Completion task: A record stores only total visitors=3. Give two different arrival schedules that produce this total. Can the total tell us the busiest hour? Explain.

Reasoning to listen for

Teacher check: Examples: three visitors at 09:00, or one each at 09:00, 10:00 and 11:00. Both total 3, but their busiest-hour patterns differ.

The total lacks arrival-time information, so the busiest hour cannot be recovered from it alone.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full **trace** only when the partial **trace** is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual reasoning check

Use the worked case for **Representation**s and information paths. Proposed explanation: “Every **output** is the final action”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

> #### Expected repair
>
> Expected correction: **Trace** an **output** entering the next stage. Ask whether the borrowing or speech stage exists and has run.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictorial practice question

Use the richer record to find blue pencils in Box B. Explain why total alone was insufficient.

Reasoned teacher answer

6–5=1 blue pencil. Total 6 alone fits many red/blue combinations, including both boxes.

Changed-case reasoning: The records become (6,2) and (6,5). Blue count can now be recovered by total minus red, assuming these are the only colours.

Collect each learner’s explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

Run the weekly evidence lab

> Run the Input, Process, Output investigation and use the first unsupported step to choose a teaching response.

Predict first, test honestly, and explain what the **evidence** means.

1. Predict

State your **prediction** before the result is known.

2. Investigate

Complete the task: **Trace** a music recommender and draw all three stages.

3. Compare cases

Keep a normal case and an edge case clearly separated.

4. Record honestly

Record what happened—even when it does not match your **prediction**.

5. Explain and plan the next test

Explain the result, one limitation, and the next useful test.

Use a fresh case after support

INDEPENDENCE After support is removed, repeat the proof with a fresh case.

Make the lab runnable

Use the concrete case as the shared normal case and the independent question as a second case. Each pair creates a boundary or missing-information case and predicts its result before testing.

Keep the expected result separate from an observed mistake.

Record	What to capture
Input or evidence	Exact data, instruction or claim
Expected result	Answer or safe fallback declared before the run
Observed result	Actual result, including errors
Explanation	Rule, calculation or evidence explaining the outcome
Next test	One justified change and a new prediction

Diagnose and reteach

Find the unsupported claim

CLAIM TO REPAIR The fan's movement is the **input**.

For guessing, return to given facts and request a **trace**. For vocabulary without application, use one example and one non-example. For arithmetic errors, check the setup before the calculation.

Finish with a different uncoached case before recording mastery.

Address these additional misconceptions

Every **output** is the final action: **Trace** an **output** entering the next stage. Ask whether the borrowing or speech stage exists and has run.

More calculation can recover discarded details: Ask students to distinguish two original records with the same retained value. No deterministic calculation of that value alone can know which original occurred.

A data type tells us what the data means: Compare a title, a summary and a full chapter. All can be text while supporting different questions.

Match the support to the first error

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

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Assess and explain the answers

> Use the Input, Process, Output questions, teacher keys and criteria after the relevant attempts. Previously discussed cases are practice; protect the separately selected

> independent task.

Worked case answer

Teacher key · Worked case answer

Input is mood; **process** is tag matching; **output** is titles. Audio playback is another step. Matching can use fixed rules; the description alone does not establish machine learning.

Independent question

Independent question · withhold the key

A classroom fan receives a temperature reading, compares it with 25 degrees C, and turns on. Name the **input**, **process** and **output**. What should happen if the reading is missing?

Expected independent answer

Teacher key · Expected independent answer

Input is the reading; **process** is comparison with 25 degrees C; **output** is the fan command. A missing reading calls for a fault signal or human review, not an invented temperature.

What counts as evidence

Criterion	0 points	1 point	2 points
Result	Incorrect or absent	Partly correct	Correct with necessary conditions
Reasoning	No supporting trace	Some steps	Clear mechanism, calculation or source support
Limit	Overclaims certainty	Vague limitation	Relevant uncertainty or missing fact
Transfer	Copies without adapting	Adapts with prompts	Solves a changed case independently

Use 6/8 as a classroom checkpoint, with 2 required for both Result and Reasoning. This is a proposed teaching rule, not a validated certification threshold.

Do not count a coached answer as independent mastery; correct unsafe or unsupported claims regardless of total.

Feedback

Ask the learner to find the **evidence** that challenges this claim: “The fan's movement is the **input**.” Keep the first attempt and an uncoached retry separate.

Original transfer answer

Teacher key · Original transfer answer

Input is the genre request, **process**ing is catalogue filtering, **output** is three titles. There is no audio-**output** stage. Filtering could use a **written rule**; the description does not establish learning.

The claim confuses title display with reading aloud.

Explanations for the additional workbook examples

W1 Trace a request

Workbook question · W1

The kiosk receives genre=science and maximum pages=120. The catalogue contains A science 80 pages, B science 150 pages and C mystery 90 pages. It keeps records with the requested genre and no more than 120 pages.

Draw the **input**, **process** and **output**, naming the surviving record.

Teacher reasoning · W1

Input includes both request fields and all three catalogue records. The **process** checks science AND pages<=120. **Output** contains A only; B fails the length condition and C fails the genre condition.

Listing A does not establish that a copy was reserved.

W2 Finish the visitor example

Workbook question · W2

A record stores only total visitors=3. Give two different arrival schedules that produce this total. Can the total tell us the busiest hour? Explain.

Teacher reasoning · W2

Examples: three visitors at 09:00, or one each at 09:00, 10:00 and 11:00. Both total 3, but their busiest-hour patterns differ. The total lacks arrival-time information, so the busiest hour cannot be recovered from it alone.

W3 Choose a sufficient representation

Workbook question · W3

For each task, choose the information needed: decide whether a book is under 120 pages; quote its first sentence exactly; decide whether a copy is available now. Explain why a genre tag alone is insufficient for all three.

Teacher reasoning · W3

Use the page count for the length condition, the actual first sentence from the correct edition for an exact quote, and a current copy-availability record for availability. Genre supplies none of those facts.

Even suitable records must be checked for the correct edition or time.

W4 Investigate a new loss of information

Workbook question · W4

A classroom device receives four scores and stores only their total, 20. A student claims that every score must be 5. Give two different lists of four nonnegative whole-number scores with total 20.

Explain precisely what the stored total establishes and what it does not.

Teacher reasoning · W4

For example [5,5,5,5] and [2,4,6,8] both total 20. The stored total establishes the sum under the stated recording **process**, but not the individual scores or their equality.

A list with fewer than four entries or a different total is not a valid counterexample.

W5 Diagnose before repairing

Workbook question · W5

A voice calculator receives a spoken request for 16 divided by 4. Its transcript says 60 divided by 4 and its calculator returns 15.

Identify the first incorrect stage, show the intended result and propose a check before calculation.

Teacher reasoning · W5

Transcription first changes 16 to 60. The calculator correctly computes the wrong supplied expression. The intended result is 4.

Display or confirm the interpreted numbers before calculating, rather than changing the arithmetic code to force the expected answer.

Assess independent transfer

Independent question · withhold the key

Selected case: A classroom device receives four scores and stores only their total, 20. A student claims that every score must be 5. Give two different lists of four nonnegative whole-number scores with total 20.

Explain precisely what the stored total establishes and what it does not.

Selected case answer

Teacher key · Selected case answer

For example [5,5,5,5] and [2,4,6,8] both total 20. The stored total establishes the sum under the stated recording **process**, but not the individual scores or their equality.

A list with fewer than four entries or a different total is not a valid counterexample.

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Teacher key · Selected case answer

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible **trace**.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Feedback that the learner uses

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

Connect and extend learning

> Connect Input, Process, Output to Helpful Classroom Sorter, adapt support and plan a later changed-case check.

Helpful Classroom Sorter

Draw the Classroom Sorter's first **input**-**process**-**output** map and show where a person may review the result.

Keep the decision, **input**, result, explanation and remaining limitation in the project **evidence** record. Retrieve this record next week rather than restarting the project.

Support and stretch

Use physical cards, spoken explanations and a three-stage drawing. Read aloud without giving the answer. Stretch: compare two **mechanism**s with the same visible **output** and explain what **evidence** distinguishes them.

Exit reflection

Ask for one decision the learner can now justify and one situation requiring more **evidence** or human review. Confidence ratings are separate from correct independent performance.

Bridge to the next week

Next: Week 3: Sensors Are Messengers. Retrieve today’s **evidence** and identify what carries forward and what is new.

Support and fresh reassessment

Provide four empty boxes and counters for learners who need arithmetic support. Let them explain **information loss** orally.

Extension: ask which extra field, such as minimum score, would rule out one of their lists and whether it would uniquely determine the original list.

W4 is secure when both lists contain four nonnegative whole numbers, both sum to 20, at least one is not all fives, and the explanation distinguishes sum from individual values.

If the learner offers only one list, request another record consistent with the same total. Use a different total for the uncoached retry.

Week 3 examines how physical signals become measurements, including the units and conditions needed to interpret recorded values.

Plan the delayed return

Suggested return points in this level: ai-1/3, ai-1/5, ai-1/8. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

Project evidence from this case

Choose sorter **input**s that retain the clues needed for its decision; draw the transformation and identify discarded details.

Prepare, teach and assess

> Use this preparation summary alongside the detailed Input, Process, Output lesson; it is a checklist rather than a replacement for the full sequence.

Teaching target

Teaching target: **Trace** information through a system without confusing information with a physical action

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. No paid AI account is required for the paper reasoning activities.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Compare the cases, Words that unlock the idea and Follow a worked example; record the brief recall in the workbook.

Session: Session 2 · Practise · 45 min

Plan: 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.

Resources: Use Find and repair a mistake, Practise together, Run the investigation and Build your understanding; use the matching workbook W2 and misconception check.

Session: Session 3 · Investigate and transfer · 45 min

Plan: 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.

Resources: Use Run the investigation, Connect your project, Show what you know, Study a complete case and Try a new case independently. Allow extra time for the full task set. Use a coding lab only when this week actually provides one; Your lesson route is the planning section.

Before class, solve the supported practice: A record stores only total visitors=3. Give two different arrival schedules that produce this total. Can the total tell us the busiest hour? Explain.

Supported practice answer

Reasoned practice answer: Examples: three visitors at 09:00, or one each at 09:00, 10:00 and 11:00. Both total 3, but their busiest-hour patterns differ.

The total lacks arrival-time information, so the busiest hour cannot be recovered from it alone.

Check every learner before discussion: Use the worked case for **Representation**s and information paths. Proposed explanation: “Every **output** is the final action”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Repair the misconception

Misconception repair: **Trace** an **output** entering the next stage. Ask whether the borrowing

or speech stage exists and has run.

Support without giving away the answer

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extend the reasoning

Extension: Store red count alongside total. Require a **prediction** and a reason before comparing. Expected result: The records become (6,2) and (6,5).

Reason: Blue count can now be recovered by total minus red, assuming these are the only colours.

Protect the independent attempt

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint

Project checkpoint: Frame the need. Name the user, the problem, allowed **input**s and a safe simple starting method.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

Use feedback, labs and repair paths

> Use feedback, earlier review and portfolios purposefully. Apply the coding routine only when this lesson has an executable lab.

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

For later coding lessons

> This week has no code block. Use this routine when a later lesson includes an executable lab.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, **trace**/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

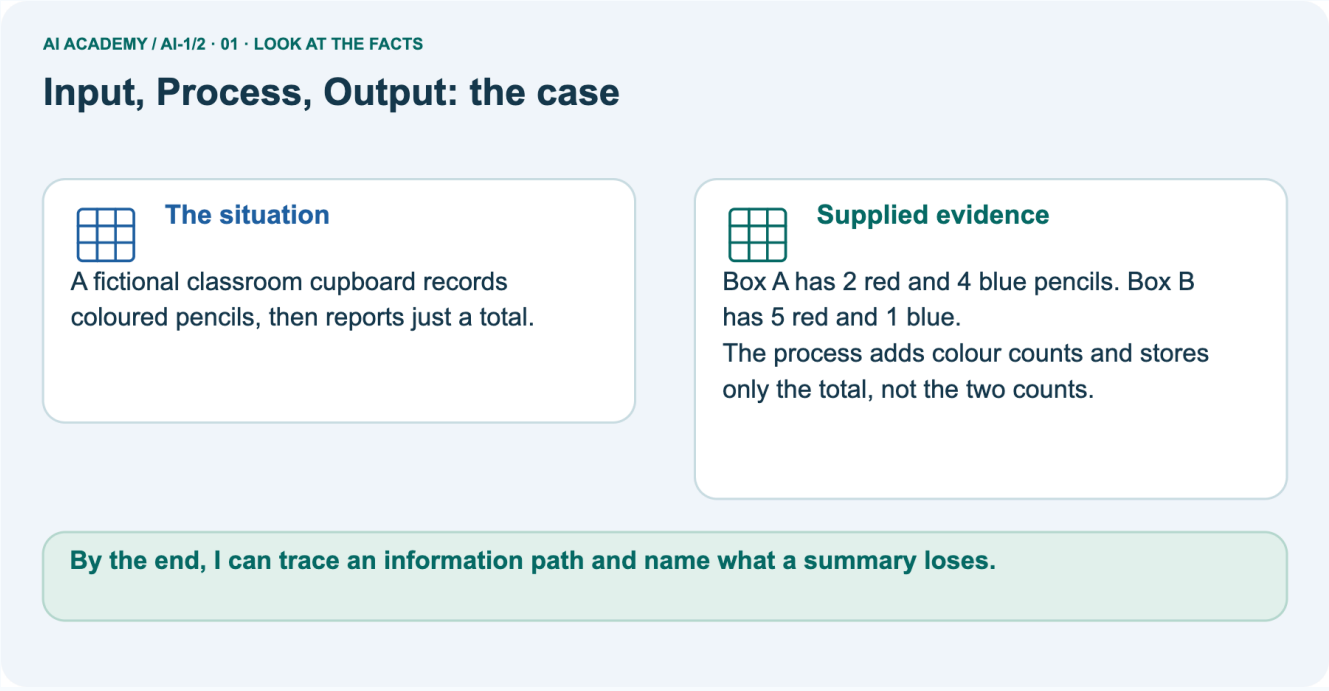
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Visual concept journey

> Show one Input, Process, Output graphic at a time, request a prediction and discuss its trace before the independent assessment.

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Input, Process, Output: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

A fictional classroom cupboard records coloured pencils, then reports just a total. Supplied facts: Box A has 2 red and 4 blue pencils. Box B has 5 red and 1 blue.

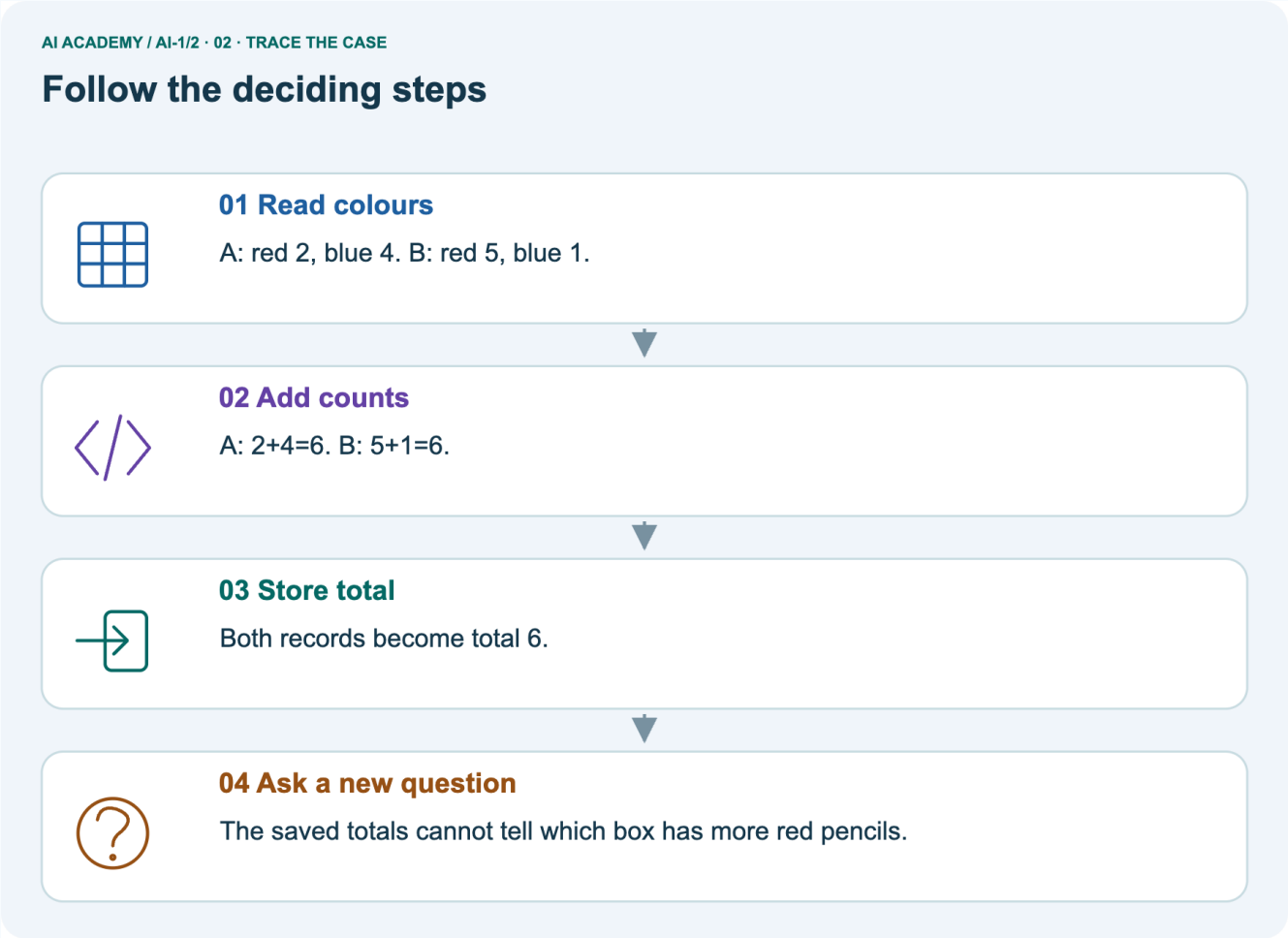
The **process** adds colour counts and stores only the total, not the two counts.

Point to the given fact that matters most. Explain why it matters.

Discuss while showing the picture

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

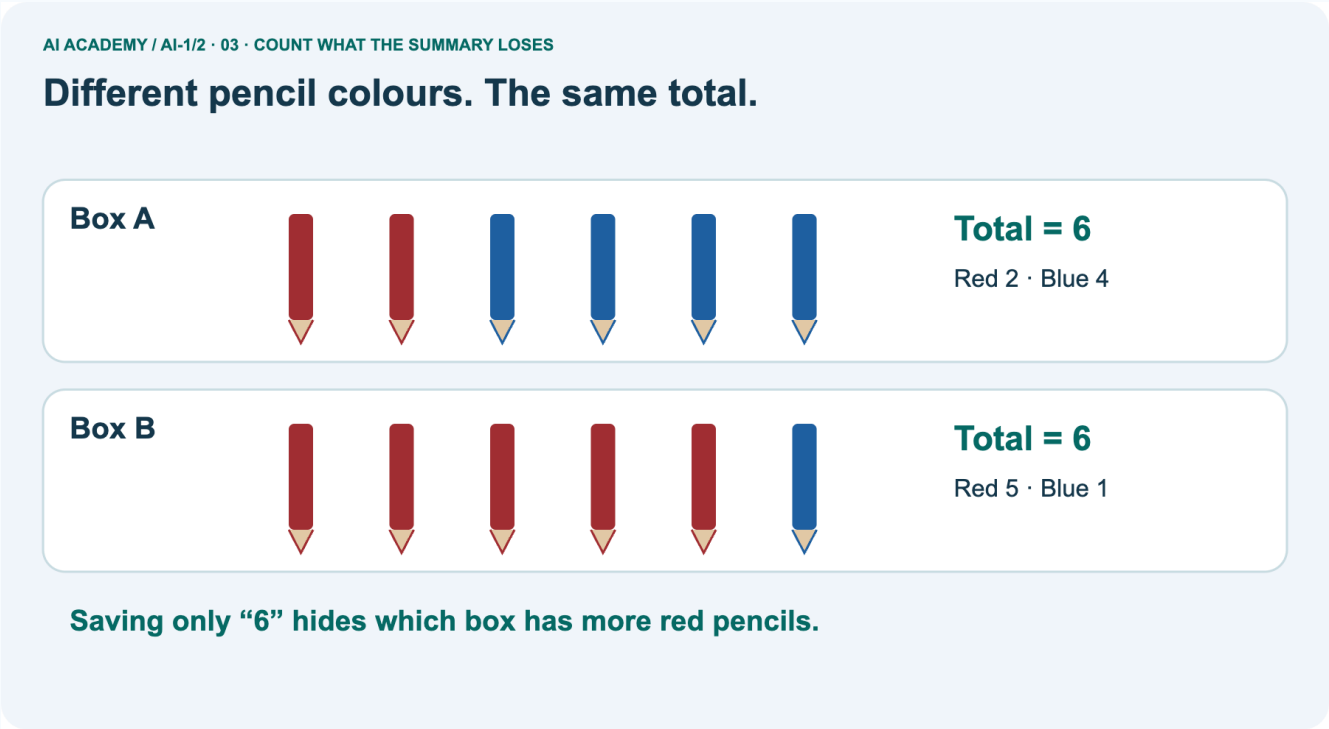
Read colours: A: red 2, blue 4. B: red 5, blue 1. → Add counts: A: 2+4=6. B: 5+1=6. → Store total: Both records become total 6. → Ask a new question: The saved totals cannot tell which box has more red pencils.

> Point to each numbered stage and explain which recorded **input** it uses.

Discuss while showing the picture

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. Different pencil colours. The same total.



Follow the labelled picture with the matching explanation. Open the vector diagram

Box A: 2 red pencils and 4 blue, total 6. Box B: 5 red pencils and 1 blue, total 6. The equal totals do not preserve the colour counts.

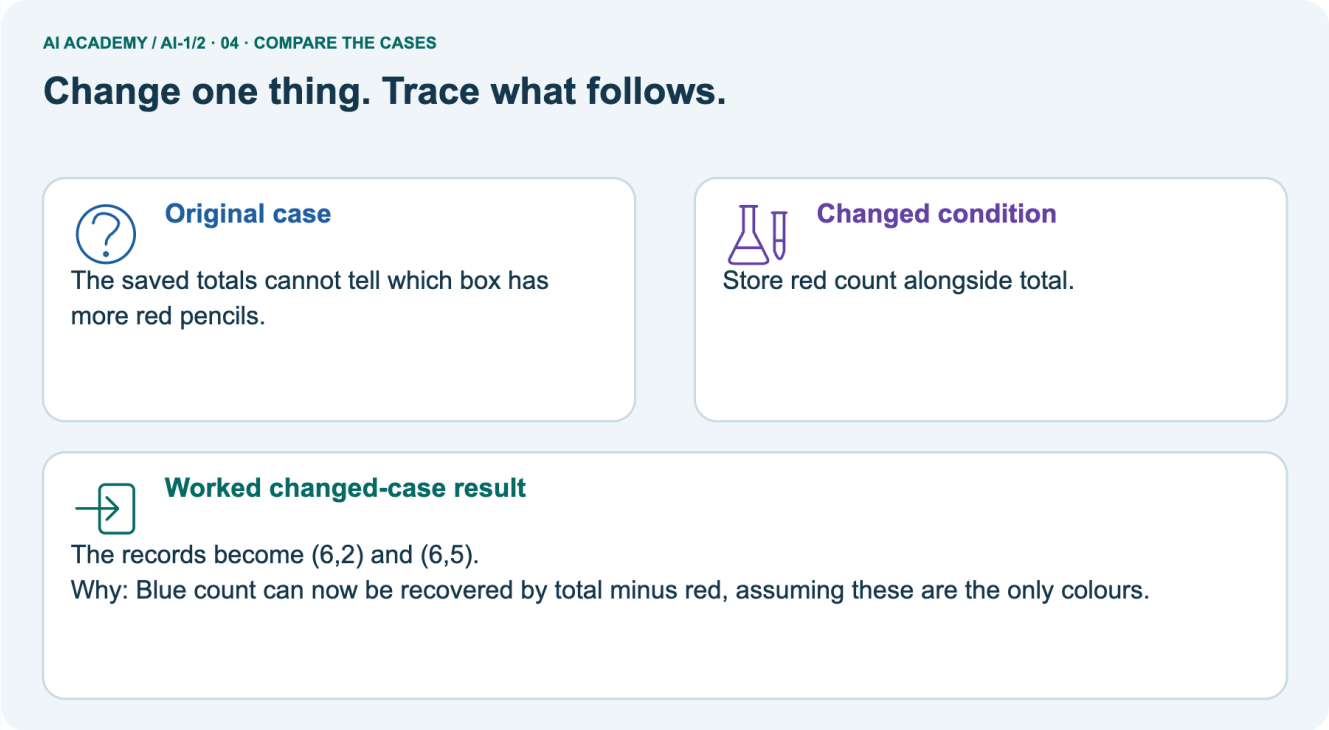
Read the labels and values. Explain the relationship the model shows.

Discuss while showing the picture

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Check row/column meanings, units and denominators before discussing a difference. Supplied toy values are not real class or model results.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: The saved totals cannot tell which box has more red pencils. Changed condition: Store red count alongside total. Expected result: The records become (6,2) and (6,5).

Reason: Blue count can now be recovered by total minus red, assuming these are the only colours.

Predict before checking the worked changed-case result. Name what changed and what the

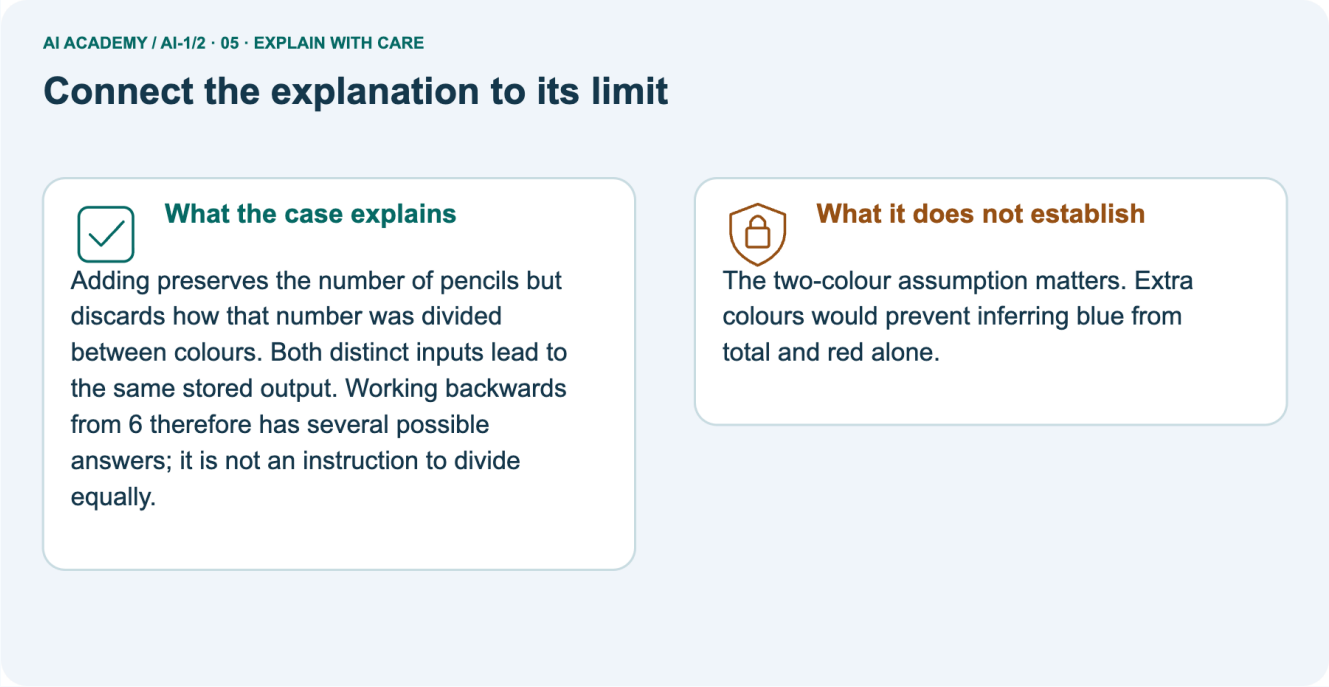
supplied facts allow you to conclude.

Discuss while showing the picture

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: Blue count can now be recovered by total minus red, assuming these are the only colours.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Adding preserves the number of pencils but discards how that number was divided between colours. Both distinct **input**s lead to the same stored **output**.

Working backwards from 6 therefore has several possible answers; it is not an instruction to divide equally. Limit: The two-colour assumption matters. Extra colours would prevent inferring blue from total and red alone.

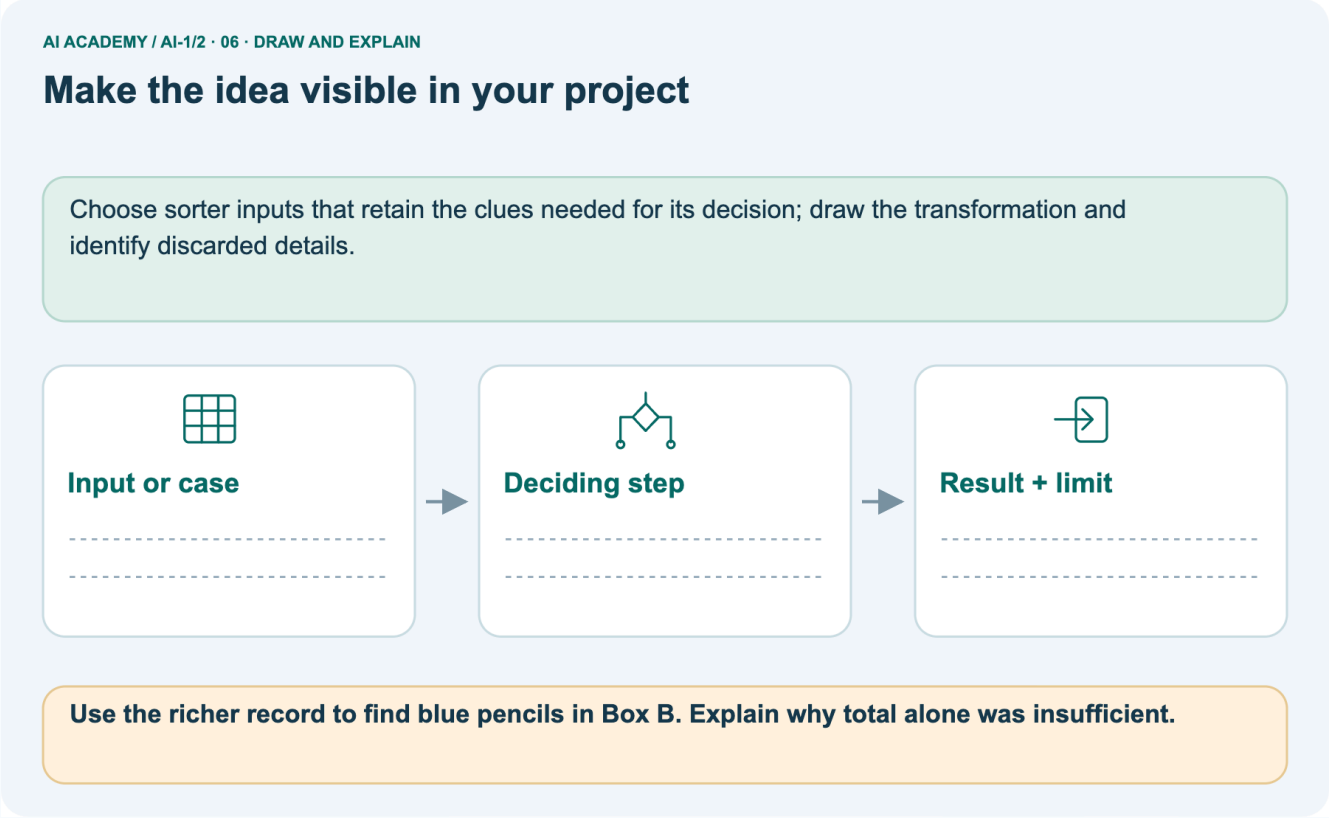
Explain the case, then point to a limit on the claim.

Discuss while showing the picture

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: The two-colour assumption matters. Extra colours would prevent inferring blue from total and red alone.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Choose sorter **input**s that retain the clues needed for its decision; draw the transformation and identify discarded details. Draw your **input**, deciding step, result and limit.

Supported question: Use the richer record to find blue pencils in Box B. Explain why total alone was insufficient.

Draw your project **trace** and justify the deciding step using **evidence**.

Discuss while showing the picture

Teacher prompt: Draw your project **trace** and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: 6–5=1 blue pencil. Total 6 alone fits many red/blue combinations, including both boxes.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.